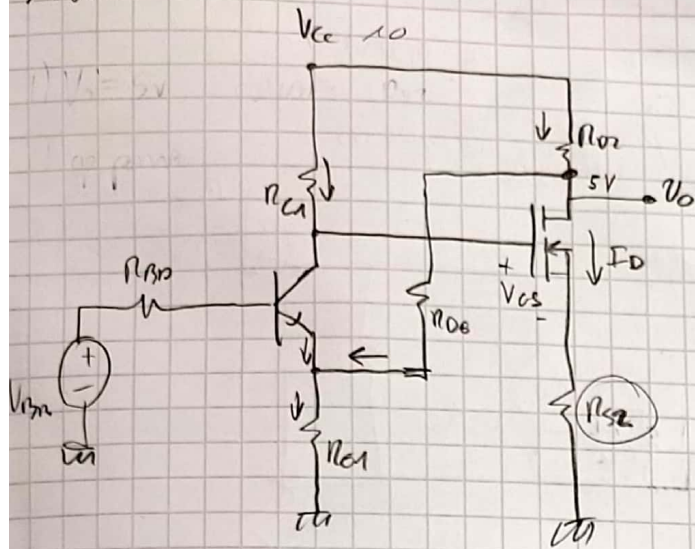


3.5.11.2023



$$V_{th} = \frac{V_{cc} R_2}{R_1 + R_2} = 3,27V$$

$$R_{th} = R_1 || R_2 = 222k\Omega$$

$$I_{D0} = \frac{V_{cc} - 5}{R_{D2}} = 0,735mA$$

$$V_{th} - R_{th} I_B - V_{BE} - R_{E1} (I_E + I_{DE}) = 0$$

$$I_{DE} = I_{D0} - I_D$$

$$V_{th} - R_{th} I_B - V_{BE} + R_{DE} I_{DE} = V_0$$

$$V_{th} - R_{th} I_B - V_{BE} - R_{E1} (1 + \beta_F) I_B - R_{E1} I_{DE} = 0$$

$$V_{th} - R_{th} I_B - V_{BE} + R_{DE} I_{DE} = V_0$$

$$I_{DE} = \frac{V_{th} - R_{th} I_B - V_{BE} - R_{E1} (1 + \beta_F) I_B}{R_{E1}}$$

$$I_B = 5,3 \mu A$$

$$I_C = 1,06mA$$

$$I_{DE} = \frac{V_0 - V_{th} + R_{th} I_B + V_{BE}}{R_{DE}}$$

$$I_{DE} = 0,328mA$$

$$I_D = I_{D0} - I_{DE} = 0,406mA$$

$$V_{GS} = 3,27V$$

$$V_{cc} - R_{D1} I_C - V_{GS} - R_{S2} I_D = 0$$

$$V_{GS1} = V_{cc} - R_{D1} I_C - R_{E1} (I_E + I_{DE}) = 6,17V$$

$$R_{S2} = \frac{V_{cc} - R_{D1} I_C - V_{GS}}{I_D} = 8k\Omega$$

$$V_{DS} = V_0 - R_{S2} I_D = 1,77V > V_{GS} - 2 = 1,27V$$

$$g_{m1} = 40,77mS^{-1}$$

$$g_{m2} = 0,636mS^{-1}$$

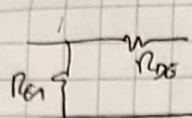
$$r_{\pi 1} = 4,09k\Omega$$

$$3) A_v = V_o / V_s$$

voltage dividing voltage sources

$$g_{m1} = 40,77 \quad V_{th} = 4,9 \text{ k}\Omega$$

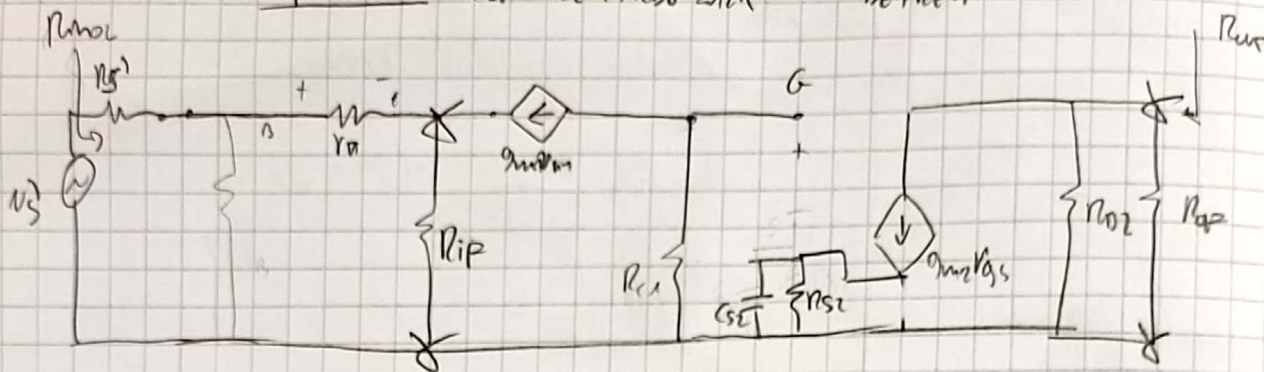
$$g_{m2} = 0,636$$



$$R_{D5} = R_{E1} // R_{D5} = 0,916 \text{ k}\Omega$$

$$R_{D5} = R_{E1} + R_{D5} = 12 \text{ k}\Omega$$

$$\beta = \frac{R_{E1}}{R_{D5} + R_{E1}} = 0,08333$$



$$R_{in2} = R_{D1} + (1 + \beta_{D1}) R_{E1} = 189 \text{ k}\Omega$$

$$R_{S1}' = R_{S1} // R_{D5} = 0,995 \text{ k}\Omega \approx R_S \text{ (theorem)}$$

$$A_{v1} = \frac{V_o}{V_s} = -g_{m2} (R_{D2} // R_{L2}) \cdot \left( \frac{-\beta_{D1} R_{E1}}{R_{S1}' + R_{in2}} \right)$$

$$= -2,76 \cdot -3,47 = 9,59$$

$$A_v^F = \frac{A_{v1}}{1 + \beta_{D1}} = 5,33 \text{ V/V}$$

$$A_v = \frac{V_o}{V_s} = A_v^F \cdot \frac{V_{S1}'}{V_s} = A_v^F \cdot \frac{R_{D5}}{R_{S1} + R_{D5}} = 5,3 \text{ V/V}$$

$$4) R_{in}^{OL} = R_{in2} + R_{S1}' = 190 \text{ k}\Omega$$

$$R_{in}^F = 361 \text{ k}\Omega$$

$$R_{in} = (R_{in}^F - R_{S1}') // R_{D5} + R_{S1} = 135 \text{ k}\Omega$$

$$5) R_{out}^{OL} = R_{D2} // R_{L2} = 4,34 \text{ k}\Omega$$

$$R_{out} = R_{E2} = 2,61 \text{ k}\Omega$$

6) f\_L due to C\_{S1} and C\_{S2}

$$R_{S1}^{SC} = R_{in}$$

$$R_{S2}^{SC} = R_{S2} // R_{L2} = R_{S2} // \frac{1}{g_{m2}} = 1,314 \text{ k}\Omega$$

→ for this feedback works  
for C\_{S2} already accounted

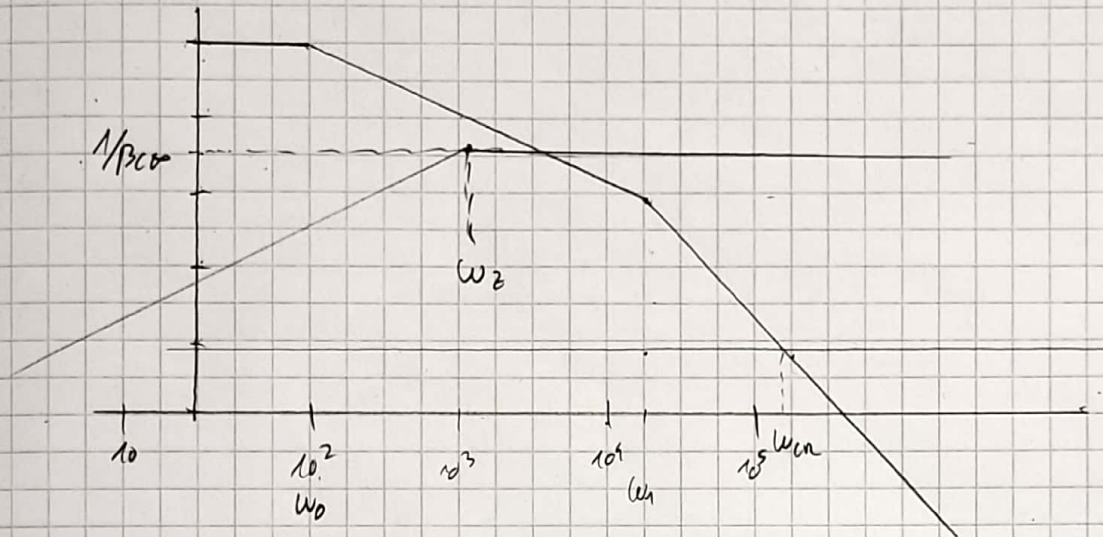
$$V_{in} = -V_{GS} \quad i_{in} = -g_{m1} V_{GS}$$

$$f_L = \frac{1}{2\pi} \left( \frac{1}{(1 + \beta_{D1}) C_{S1} R_{S1}^{SC}} + \frac{1}{C_{S2} R_{S2}^{SC}} \right) = 18,52 \text{ Hz}$$

BS 2)

$$1) A(s) = 1 + \frac{R_2}{R_1} = 3,06$$

$$2) T = \frac{W}{v_f} = A(s) \cdot \frac{R_1}{R_2 + R_1} \quad \beta_0 = \beta_{\infty} = 0,326 \quad \frac{1}{\beta_{\infty}} = 3,7 \text{ dB}$$



$$A_{d0} \omega_0 = 1,1 \omega_{in}$$

$$A_{d0} \omega_0^2 = A/\beta_{\infty} \omega_{in}^2 \quad \omega_{in} = \sqrt{A_{d0} \omega_0 \omega_{in} \beta_{\infty}} = 2,55 \cdot 10^5 \text{ rad/s} \Rightarrow \boxed{\beta_{\infty} = 40,7 \text{ kHz}}$$

$$\varphi_m = 180 - 2 \arctan\left(\frac{\omega_{in}}{\omega_0}\right) - \arctan\left(\frac{\omega_{in}}{\omega_m}\right) = \boxed{4,5^\circ}$$

3) we introduce

$$\omega_z = \frac{1}{C R_C} = 1,377 \cdot 10^3 \text{ rad/s}$$

$$\omega_p = \frac{1}{C(R_C + (R_1 || R_2))} = 1,36 \text{ rad/s}$$

$$\left(\beta_{\infty}\right)^{-1} = \left(\frac{R_1 || R_C}{R_2 + R_1 || R_C}\right)^{-1} = 3090 = 70 \text{ dB}$$

$$A_{d0} \omega_0 = 1/\beta_{\infty} \omega_{in} R_C \rightarrow \omega_{in} R_C = A_{d0} \omega_0 \beta_{\infty} = 3,235 \cdot 10^3 \text{ rad/s} \Rightarrow \beta_{\infty} = \boxed{515 \text{ kHz}}$$

$$\varphi_m = 180 - 2 \arctan\left(\frac{\omega_{in}}{\omega_0}\right) - \arctan\left(\frac{\omega_{in}}{\omega_m}\right) - \arctan\left(\frac{\omega_{in}}{\omega_p}\right) + \arctan\left(\frac{\omega_{in}}{\omega_z}\right) = \boxed{59,55^\circ}$$

4) ideal is when  $A(s) \rightarrow \infty$

$$A_v = \frac{1}{\beta(s)} = \frac{R_2 + R_1 || \frac{1 + s C R_C}{s C}}{R_1 || \frac{1 + s C R_C}{s C}} =$$